

Branded Short-Link & Bio-Link Hub

Bitly + Linktree Hybrid

🔒 Universal Security & Architecture Requirements

- **Pair Token Auth:** Short-lived JWT Access Token (15m) + Long-lived Refresh Token (7d) in `httpOnly` cookies.
- **Auth Features:** Signup with Email Verification simulation, Login with token rotation, Forgot/Reset Password.
- **UI Standard:** Build all interface components using coss.com/ui primitives.

1. Overview & Goal

Build a high-performance URL shortening engine with custom vanity slugs and click telemetry analytics, alongside a customizable "Link-in-Bio" hub manager for creator profile pages.

2. Technical Feature Specifications

A. High-Speed URL Redirection Engine

- Auto-generates unique 6-character short codes or accepts custom vanity slugs (e.g., `/r/summer-sale`).
- Duplicate alias collision detection and valid URL format enforcement.
- Redirection endpoint (`GET /r/:shortCode`): Returns `302 Found` and logs click events asynchronously.

B. Click Analytics & Metrics Aggregation

- Captures click metadata: timestamp, HTTP referrer, device type (*Mobile, Desktop, Tablet*), and IP hash.
- Dashboard charts: Total clicks over time, top referrers, and device distribution metrics.

C. Link Library Studio

- Management table showing destination URL, short link, one-click copy button, QR code generation, and delete actions.
- Search and pagination across user's link library.

D. "Link-in-Bio" Hub Customizer & Public Page

- Visual builder to configure avatar, `displayName`, `bio`, and manage social link buttons.
- Theme selector (*Minimal Light, Dark Slate, Gradient*).
- Public mobile-responsive route (`/bio/:username`) rendering the configured profile page.

E. Rate Limiting & Security

- Rate limiting (`express-rate-limit`) on creation and redirect routes to prevent abuse.

3. Deliverables Checklist

Area	Deliverable Criteria
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Auth & Security	Pair token JWT auth, route protection, rate limiting configuration.
Redirection & Data	Indexed slug lookups, click telemetry schema, analytics aggregations.
Frontend	Coss UI components, QR code generator, theme switcher, mobile bio page.
Repository	Clean GitHub repo, <code>.env.example</code> , API documentation.